🔑 **Answer Key — Teacher Copy · Lesson 9-1**

MSTAR PRACTICE WORKSHEET

Lesson 9-1: Explore Relationships Between Two Variables

Grade 6 Mathematics · Standard 6.AT.11

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

Akela chooses how many hours of sunlight each tray gets. Which is the INDEPENDENT variable?

A. Plant height

B. Hours of sunlight

C. Number of trays

D. Number of seeds

Akela chooses the sunlight, so sunlight is subject to choice — that makes it independent.

Question 2 SELECTED RESPONSE — CORRECT: D

In Akela's experiment, which is the DEPENDENT variable?

A. Hours of sunlight

B. The kind of seed

C. The size of the tray

D. Plant height

Height responds to how much sunlight the tray received, so height is the dependent variable.

Question 3 SELECTED RESPONSE — CORRECT: C

Claire burned 123 Calories on a 30-minute jog. Which quantity is subject to choice?

A Calories she burns

B Her heart rate

C Minutes she jogs

D The weather

> Claire decides how long to jog. The Calories follow from that decision.

Question 4 SELECTED RESPONSE — CORRECT: D

Which sentence is written in the correct order?

A Minutes jogged depends on Calories burned.

B Calories burned depends on Calories burned.

C Neither quantity depends on the other.

D Calories burned depends on minutes jogged.

> The dependent quantity comes first in the sentence '___ depends on ___ .' Calories respond to time, so Calories depend on minutes.

Question 5 SELECTED RESPONSE — CORRECT: C

A tray that got 2 hours of sun grew to 4 cm. A tray that got 6 hours grew to 12 cm. As sunlight increases, plant height...

A decreases

B stays the same

C increases

D cannot be determined

> 2 hours gave 4 cm and 6 hours gave 12 cm, so more sunlight went with more height.

Question 6 SELECTED RESPONSE — CORRECT: A

You buy tickets at $45 each. Which quantity depends on the other?

- (A) Total cost depends on number of tickets
- (B) Number of tickets depends on total cost
- (C) The price per ticket depends on the total cost
- (D) Nothing depends on anything here

You choose how many tickets to buy; the total cost is the result of that choice.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

Devon adds a different number of scoops of plant food to each tray of seedlings. Two weeks later he counts the new leaves on every tray. Which pair correctly names the variables?

A. Independent: new leaves counted; Dependent: scoops of plant food

B. Independent: scoops of plant food; Dependent: new leaves counted

C. Independent: the number of trays; Dependent: scoops of plant food

D. Independent: new leaves counted; Dependent: the two weeks of growing time

> Devon chooses how many scoops of plant food a tray receives, so the scoops are subject to choice and are the independent variable. The new leaves respond to that choice, so the leaf count is the dependent variable. The sentence check works only one way: the number of new leaves depends on the scoops of plant food.

- **A:** These are reversed. Devon chooses how many scoops a tray gets, and the leaves respond.
- **C:** The trays are only containers. What Devon varies on purpose is the scoops of plant food.
- **D:** The two weeks are the same for every tray, so nothing there varies. Devon sets the scoops.

Part B — correct answer: C

Which sentence shows why the pair chosen in Part A is correct?

A. The scoops of plant food depend on the number of new leaves, because Devon counts the leaves last.

B. The scoops are independent because the number of scoops is the larger number.

C. The number of new leaves depends on the scoops of plant food, because Devon chooses the scoops and the leaves respond.

D. The number of new leaves is independent because it is the quantity the question asks about.

> The sentence check reads in only one direction. The leaves respond to the plant food, so the leaf count depends on the scoops, and the quantity that is depended on, the scoops, is the independent variable. The order in which Devon measures, the size of the numbers, and what the question asks about do not decide which variable is independent.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, C, E

Select ALL statements that name the independent variable correctly for their own situation.

- **A** A gardener chooses how many minutes to run the sprinkler, so the minutes the sprinkler runs is the independent variable.
- **B** A student hangs different numbers of weights on a spring, so the stretch of the spring is the independent variable.
- **C** Claire decides how long to jog, so the minutes she jogs is the independent variable.
- **D** The total cost of the tickets is the independent variable, because the cost is the amount the problem asks for.
- **E** A tray receives a set number of hours of sunlight each day, so the hours of sunlight is the independent variable and the plant height depends on it.

> In each true statement the quantity named is the one that is chosen, and the other quantity responds to it. In the spring statement the student chooses the weights, so the weights are independent and the stretch is dependent. In the ticket statement the number of tickets is chosen and the cost responds, so the cost is dependent no matter what the problem asks for.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.